CURTIS LEE

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EDUCATION

California Institute of Technology (Caltech) — Pasadena, CA **2024 – 2028**
B.S. in Chemical Engineering (Process track) and Business, Economics, and Management — GPA 3.8 / 4.0

The King's School Canterbury — Kent, UK **2019 – 2024**
A-Levels: Mathematics, Further Mathematics, Physics, Chemistry

- Gold awards: RSC UK Chemistry Olympiad (2024), Cambridge Chemistry Challenge (2023), BPhO Senior Physics Challenge (2023). School Monitor; Deputy Concertmaster of the school orchestra; composed and conducted a work for a 50-person student choir.

EXPERIENCE

The Hong Kong and China Gas Company (Towngas) — Green Fuels & Chemicals **Jun – Aug 2026**
Process Engineering Intern — Foshan, Guangdong & Jungar Banner, Inner Mongolia, China

- Authored a pre-FEED design report converting bauxite residue and soybean processing residue into light olefins via supercritical water gasification, bi-reforming, and OXZEO synthesis; closed mass and carbon balances across ten 300 t/d hydrothermal trains (42–55 kt/y olefins). curtis-lee.com/projects/supercritical-water-gasification
- Built a 20-year techno-economic model returning ~3% IRR against a 10% hurdle, and advised against commercial FEED on those terms; modelled four alternative cases, finding carbon recovery a more durable lever than a certified export premium.
- Supported engineering on a new green-fuels plant in Foshan, using the operating Jungar Banner green-methanol plant as a reference case to ground the design in how a plant runs.

UC Davis — Robert Mondavi Institute, iCAMP (Alternative Meat & Protein) **Jul – Aug 2025**
Cultivated Meat Research Intern — Davis, CA

- Screened fetal-bovine-serum-free growth media to cut the cost and animal dependence of cultivated meat.
- Built and tested 32 scaffold combinations, using Texture Profile Analysis to benchmark against real meat texture.

National University of Singapore — Institute for Functional Intelligent Materials **Mar – Apr 2024**
Research Intern — Singapore (invited by Prof. K. Novoselov, 2010 Nobel Laureate in Physics)

- Grew graphene by Chemical Vapour Deposition under Dr. Maxim Rybin, tuning growth conditions and characterizing films.

cook.enterprise — Young Enterprise UK Competition **Nov 2022 – Apr 2023**
General Manager — Kent, UK

- Selected to lead a 17-person team; set direction across product, budget, and production, delivered a cookbook earning £5,850 in revenue, and won the Kent Finals Best Company award. Cookbook at curtis-lee.com/projects/cook-enterprise

Caltech — Cooking Class (Instr. Tom Mannion) **Sep 2024 – Present**
Teaching Assistant — Pasadena, CA

- Teach students to cook through weekly practicals.

Michelin-Starred Restaurants — Stages **2021 – Present**
The Fat Duck • Core by Clare Smyth • The Clove Club • Écriture (HK) — chefs totalling 42 stars

- Worked alongside some of the best chefs in high-pressure, zero-error kitchens.

PROJECTS

- Long/short equity strategy (BEM 114): built a market-neutral, event-driven strategy scoring earnings-call transcripts with FinBERT sentiment, long the weakest-sentiment names and short the strongest, controlling for Fama–French factors. Held near-zero beta and produced a statistically significant 1.18% monthly alpha ($p = 0.022$).
- Biodiesel from used cooking oil (ChE62 & Ch9): designed and ran a two-stage process converting waste cooking oil to biodiesel, using acid-catalyzed esterification to reduce free fatty acids before base-catalyzed transesterification, and substituting bio-derived ethanol for fossil methanol to yield fatty acid ethyl esters at 75%+ conversion.

SKILLS & LANGUAGES

- **Technical:** Python, JavaScript/TypeScript, React/Next.js; NLP (FinBERT), factor modelling; process design, techno-economic and mass/energy balance modelling, DWSIM; NMR, IR, mass spectrometry, Texture Profile Analysis, cell culture, CVD. Built engineering modelling and planning tools (curtis-lee.com/tools).
- **Languages:** English, Cantonese, Mandarin (native); French (elementary).